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Market Basket Analysis and Intelligent Product Recommendation using Association Rule Mining and MLOps

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Contents

1	Introduction	1
1.1	Project Objective	1
1.2	Importance of MLOps in Data Science	2
1.3	Project Structure and Report Overview	2
2	Chosen MLOps Platform: Weights & Biases	4
2.1	Platform Overview	4
2.2	Key Features and Capabilities of W&B	4
2.3	Comparison with Other MLOps Platforms	5
3	MLOps Architecture with Weights & Biases	6
3.1	Architecture Description	6
3.2	End-to-End Workflow	7
3.3	Experimentation and Pipeline Management	8
3.3.1	Experiment Tracking with W&B	8
3.3.2	W&B Artifacts for Data and Model Versioning	8
3.3.3	Hyperparameter Sweeps and Automation	9
3.4	How W&B Supports MLOps Principles	10
4	Experimentation and Model Development	11
4.1	Description of the Machine Learning Task Used	11
4.2	Steps to Implement and Track the Model Using W&B	12
4.2.1	Data Preparation and Artifact Creation	12
4.2.2	Experimental Design and W&B Run Tracking	12
4.2.3	Model Selection and Artifact Promotion	13
4.3	Results and Observations	13
4.3.1	Experiment Tracking and Visualization	13
4.3.2	Algorithm Performance Analysis	14
4.3.3	Key Observations and Insights	15

5	Challenges and Limitations	16
5.1	Challenges Faced During Experimentation with W&B	16
5.1.1	Integration with Kaggle Environment	16
5.1.2	Learning Curve for MLOps Concepts	16
5.1.3	Algorithm-Scale Mismatch	17
5.2	Limitations of the W&B Platform	17
5.2.1	Artifact Management for Non-Traditional Models	17
5.2.2	Offline-First Workflow Constraints	18
5.2.3	Cost and Accessibility Considerations	18
5.3	Limitations of the Association Rule Mining Approach	18
5.3.1	Data Limitations	18
5.3.2	Algorithmic Limitations	18
5.4	Future Work Directions	19
5.4.1	MLOps Enhancements	19
5.4.2	Analytical Extensions	19
5.4.3	Production Considerations	19
6	Conclusion	20
6.1	Key Findings and Outcomes	20
6.1.1	MLOps Enhances Reproducibility and Decision-Making	20
6.1.2	Systematic Algorithm Comparison Reveals Clear Trade-offs	20
6.1.3	Interpretable Outputs with Direct Business Application	21
6.2	Practical Implications	21
6.2.1	For Data Mining Education	21
6.2.2	For Retail Analytics	21
6.2.3	For MLOps Adoption	21
6.3	Final Reflections	22
	References	23
	GitHub Repository	24

List of Figures

3.1	MLOps architecture showing the integration of W&B with the market basket analysis workflow. The platform manages experiment tracking, artifact versioning, and model selection.	6
3.2	Artifact lineage in W&B showing dependencies between data, experiments, and model artifacts.	9
4.1	W&B runs table showing all Apriori and FP-Growth experiments. Each row represents a unique configuration with logged hyperparameters (MIN_S, MIN_C) and performance metrics (RUNTI, MIN_L). The table enables systematic comparison and selection of the best-performing model.	13
4.2	W&B experiment dashboard showing visual analysis of hyperparameter configurations and performance metrics for all Apriori and FP-Growth runs.	14

List of Tables

3.1	Mapping W&B features to MLOps principles in our project	10
4.1	Consolidated results showing key performance metrics for selected configurations.	14

Abstract

This project explores the implementation of MLOps practices using Weights & Biases (W&B) through a practical case study in market basket analysis. Using the Instacart dataset, we systematically compare two association rule mining algorithms—Apriori and FP-Growth—within a structured experiment tracking framework. The workflow demonstrates how MLOps principles enhance reproducibility and decision-making in data mining projects: data preprocessing outputs are versioned as artifacts, algorithm configurations are tracked as experiments, and model selection follows quantitative criteria evaluated through the W&B dashboard. The resulting rule-based recommendation system shows how interpretable patterns can drive business decisions, while the integrated MLOps approach ensures that every analytical step is documented, versioned, and reproducible. This work illustrates how traditional data mining techniques benefit from modern ML infrastructure, providing a template for applying MLOps to unsupervised learning tasks.

Chapter 1

Introduction

1.1 Project Objective

The primary objective of this project is to gain hands-on experience with **Weights & Biases (W&B)** as an MLOps platform by implementing a complete market basket analysis pipeline. Through this exploration, we aim to understand how W&B facilitates the implementation of MLOps principles—reproducibility, collaboration, and scalability—in a practical data mining context. The Instacart Market Basket Analysis dataset serves as our concrete case study, allowing us to demonstrate how association rule mining can be systematically managed within a modern MLOps workflow.

Specifically, this project aims to:

- Design and implement an end-to-end pipeline for market basket analysis, from raw transactional data to actionable product recommendations.
- Compare two classical association rule mining algorithms—Apriori and FP-Growth—using systematic experiment tracking within W&B.
- Demonstrate how MLOps practices, including artifact versioning and experiment comparison, enhance the reproducibility and decision-making process in data mining projects.
- Build a rule-based recommendation system that showcases the practical business value of interpretable association rules.
- Provide a template for applying MLOps principles to unsupervised learning tasks traditionally managed through ad-hoc workflows.

1.2 Importance of MLOps in Data Science

In the rapidly evolving field of data science and machine learning, the need for robust frameworks to manage the end-to-end lifecycle of analytical projects has become critical. **MLOps** (Machine Learning Operations) bridges the gap between data science experimentation and engineering deployment by providing standardized practices and tools for versioning, tracking, and reproducibility.

Traditional market basket analysis often involves iterative experimentation with different algorithms and parameters, but without systematic tracking, it becomes difficult to reproduce results, compare configurations, or maintain clear lineage from data to insights. MLOps platforms like Weights & Biases address these challenges by:

- Providing centralized experiment tracking to log parameters, metrics, and outputs.
- Enabling version control for datasets, preprocessing steps, and model artifacts.
- Facilitating collaboration through shared projects and visual comparison tools.
- Ensuring scalability from small exploratory projects to production-grade systems.

This project illustrates how MLOps transforms association rule mining from a one-off analysis into a reproducible, auditable workflow where every decision—from data filtering to algorithm selection—is documented and versioned.

1.3 Project Structure and Report Overview

This report is structured to reflect both the technical implementation of market basket analysis and the MLOps framework that organizes it. Following this introduction:

- **Chapter 2** introduces Weights & Biases as our chosen MLOps platform, comparing it with alternatives and justifying our selection.
- **Chapter 3** details the MLOps architecture implemented with W&B, describing the end-to-end workflow and how the platform supports key principles.
- **Chapter 4** presents our experimentation with association rule mining, showing how Apriori and FP-Growth were systematically evaluated using W&B’s tracking capabilities.
- **Chapter 5** discusses challenges encountered and limitations of both our approach and the platform.
- **Chapter 6** concludes with key findings and implications for both data mining practice and MLOps adoption.

By framing market basket analysis within an MLOps context, this project demonstrates how traditional data mining techniques benefit from modern engineering practices, resulting in more reliable, scalable, and actionable insights.

Chapter 2

Chosen MLOps Platform: Weights & Biases

2.1 Platform Overview

Weights & Biases (W&B) is a lightweight MLOps platform designed to help data science teams track experiments, version datasets and models, and collaborate around machine learning projects. It provides a central place where runs, configurations, metrics, and artifacts can be logged and compared, which is particularly valuable when exploring multiple algorithms or hyperparameter settings. In this project, W&B is used to structure the market basket analysis pipeline built on the Instacart dataset and to make the association-rule experiments more transparent and reproducible.

2.2 Key Features and Capabilities of W&B

Among the many features offered by W&B, the following are most relevant for this mini project:

- **Experiment tracking:** logging configuration dictionaries (algorithm, support, confidence, lift) together with scalar metrics such as runtime, number of rules, average confidence, and lift for each mining run.
- **Artifact management:** storing the preprocessed transaction matrix and the selected association-rule tables as versioned artifacts that can be reused across notebooks and re-downloaded to reproduce experiments.
- **Projects and reports:** grouping runs and artifacts into a single project dedicated to Instacart market basket analysis, which simplifies comparison between Apriori and FP-Growth and supports concise reporting.

These capabilities are sufficient to support a disciplined, MLOps-inspired workflow even for a relatively small-scale, association-rule-based recommender.

2.3 Comparison with Other MLOps Platforms

Several alternative MLOps platforms exist, including MLflow, Kubeflow, and cloud-specific solutions such as Amazon SageMaker or Vertex AI. MLflow, for example, also provides experiment tracking and model packaging, but typically requires more setup around tracking servers or integrations with other tools. Kubeflow focuses on orchestrating complex, container-based pipelines on Kubernetes clusters, while managed services offer tight integration with specific cloud ecosystems.

In contrast, W&B offers a simple, hosted interface that can be integrated directly into Kaggle notebooks and other Python environments with minimal configuration. This makes it well suited for a teaching-oriented project such as this one, where the emphasis is on understanding the role of MLOps in a data mining workflow rather than building large, production-grade deployment stacks.

Chapter 3

MLOps Architecture with Weights & Biases

3.1 Architecture Description

The architecture implemented in this project centers Weights & Biases (W&B) as the orchestration layer for the entire association rule mining workflow. This design transforms a traditional data mining pipeline into a reproducible, version-controlled MLOps system. As illustrated in Figure 3.1, the architecture consists of three interconnected layers:

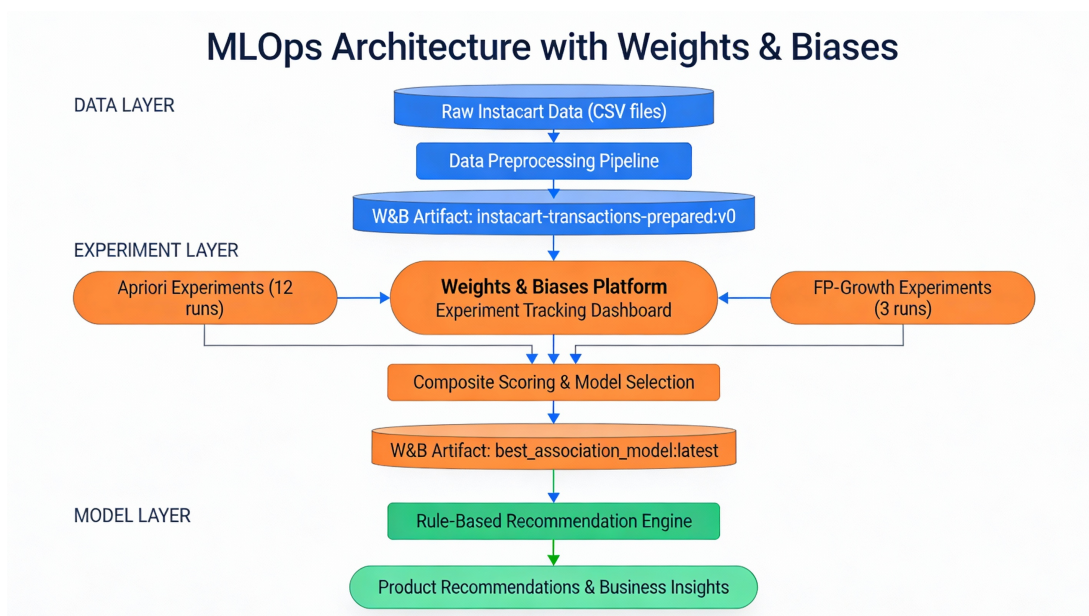


Figure 3.1: MLOps architecture showing the integration of W&B with the market basket analysis workflow. The platform manages experiment tracking, artifact versioning, and model selection.

1. **Data Layer:** Raw Instacart tables are processed into a transactional format, with the resulting matrix versioned as a W&B artifact.

2. **Experiment Layer:** Multiple association rule mining configurations (Apriori and FP-Growth with varying hyperparameters) are executed as tracked W&B runs.
3. **Model Layer:** Selected rule sets are packaged as model artifacts and made available for deployment in recommendation systems.

W&B serves as the unifying platform that connects these layers, providing a single source of truth for all experiments, data versions, and model outputs.

3.2 End-to-End Workflow

The workflow implemented in this project follows a systematic progression from raw data to actionable insights, with W&B ensuring traceability at each stage:

1. **Data Preprocessing and Artifact Creation:** The Instacart dataset is loaded, cleaned, and transformed into a transaction-item matrix. This matrix is logged to W&B as the `instacart-transactions-prepared` artifact, establishing a versioned baseline for all subsequent experiments.
2. **Association Rule Experimentation:** Using the versioned transaction artifact, multiple mining configurations are executed. Each combination of algorithm (Apriori/FP-Growth) and hyperparameters (support, confidence) runs as an independent W&B experiment, with metrics automatically logged for comparison.
3. **Systematic Evaluation and Comparison:** All experiments are aggregated within the W&B dashboard. A composite scoring function evaluates each configuration based on rule quality (lift, confidence), coverage (number of rules), and computational efficiency (runtime).
4. **Model Selection and Packaging:** The highest-scoring configuration is identified through the W&B interface. Its rule set is promoted to a `best_association_model` artifact, creating a clear lineage from experiment to deployed model.
5. **Application and Deployment:** The selected model artifact is retrieved by the recommendation module, which uses the association rules to generate product suggestions based on user baskets.

This workflow embodies core MLOps principles while remaining accessible for a market basket analysis context, demonstrating how even traditional data mining tasks benefit from modern ML infrastructure.

3.3 Experimentation and Pipeline Management

3.3.1 Experiment Tracking with W&B

W&B’s experiment tracking system provided the foundation for systematic comparison of mining algorithms. Each experimental run was structured as follows:

- **Configuration:** Algorithm choice (Apriori or FP-Growth), hyperparameters (min_support, min_confidence, min_lift), and experiment metadata.
- **Metrics:** Runtime, number of frequent itemsets, number of rules, average confidence, average lift, maximum lift.
- **Artifacts:** Generated rule CSV files attached to each run for complete reproducibility.

The tracking implementation used W&B’s Python SDK:

```
import wandb

# Initialize a run
run = wandb.init(project="instacart-market-basket",
                 config={"algorithm": "apriori",
                        "min_support": 0.002,
                        "min_confidence": 0.3})

# Log metrics
run.log({"runtime": 4.35,
        "num_rules": 54,
        "avg_lift": 11.49})
```

This structured approach transformed ad-hoc algorithm testing into a systematic exploration where every configuration change was documented and comparable.

3.3.2 W&B Artifacts for Data and Model Versioning

Artifact management in W&B created a reproducible lineage throughout our pipeline:

- **Data Artifacts:** The preprocessed transaction matrix was logged as artifact `instacart-transactions-prepared:v0`. Downstream experiments explicitly reference this version, ensuring consistent inputs.
- **Model Artifacts:** Each experimental run produced rule sets as CSV files. The best-performing configuration’s rules were promoted to `best_association_model:latest`.

- **Lineage Tracking:** W&B automatically tracks dependencies between artifacts. As shown in Figure 3.2, the final model artifact traces back through specific experiments to the original data version.

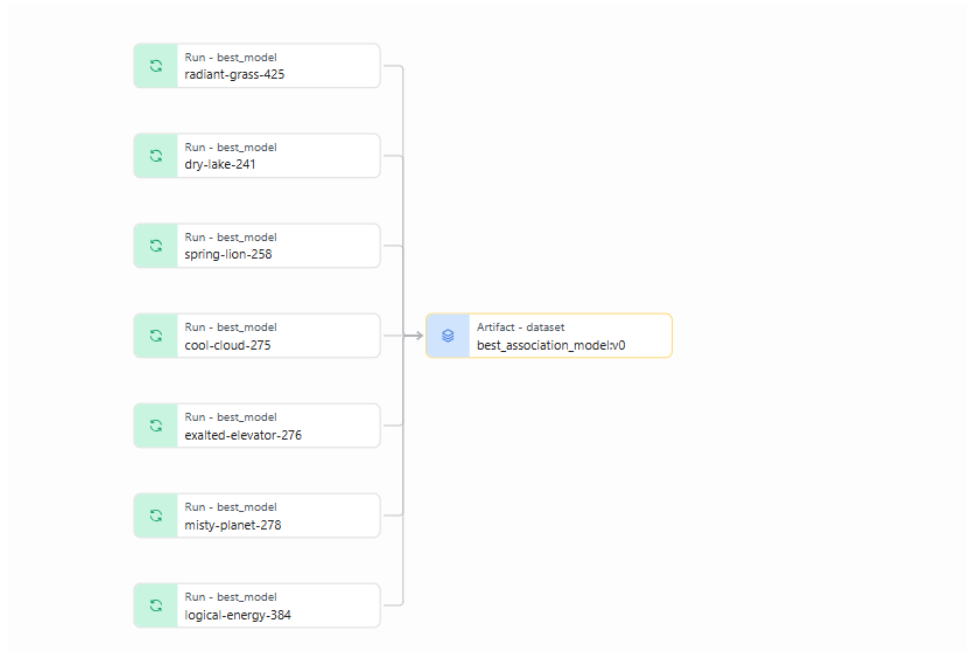


Figure 3.2: Artifact lineage in W&B showing dependencies between data, experiments, and model artifacts.

3.3.3 Hyperparameter Sweeps and Automation

While this project used manually defined parameter grids, the architecture supports full automation through W&B's sweep functionality. A sweep configuration could automatically explore the hyperparameter space:

```
sweep_config = {
    "method": "grid",
    "parameters": {
        "algorithm": {"values": ["apriori", "fpgrowth"]},
        "min_support": {"values": [0.001, 0.002, 0.005, 0.01]},
        "min_confidence": {"values": [0.1, 0.2, 0.3]}
    }
}
```

This capability positions the workflow for scalability: as the transaction dataset grows or additional algorithms are introduced, W&B can manage the combinatorial explosion of configurations automatically.

3.4 How W&B Supports MLOps Principles

The implemented architecture demonstrates how W&B enables key MLOps principles in practice:

Table 3.1: Mapping W&B features to MLOps principles in our project

MLOps Principle	W&B Feature Utilized	Project Implementation
Reproducibility	Artifact versioning and dependency tracking	All experiments reference specific data artifact versions; complete environment captured
Comparability	Unified experiment dashboard and visualization	Apriori and FP-Growth runs compared in parallel coordinates charts
Collaboration	Shared projects and run commenting	Project accessible to team members for review and discussion
Scalability	Cloud-based infrastructure and sweep automation	Architecture supports expanding to larger datasets and parameter spaces
Deployment Readiness	Model registry and artifact packaging	Best model artifact can be fetched by production recommendation service

The architecture successfully translates abstract MLOps concepts into concrete practices for association rule mining. By treating transaction matrices and rule sets as versioned artifacts, and mining configurations as tracked experiments, we bridge the gap between exploratory data mining and production-ready ML systems.

Chapter 4

Experimentation and Model Development

4.1 Description of the Machine Learning Task Used

This project focuses on **Market Basket Analysis (MBA)** as the core machine learning task. MBA is an unsupervised learning technique that discovers relationships between products in transactional data, identifying patterns of the form "customers who buy items in set A also tend to buy items in set B ". These patterns, known as association rules, provide interpretable insights that can directly inform business strategies such as product recommendations, cross-selling, and store layout optimization.

For this task, we evaluated two classical algorithms with different computational characteristics:

- **Apriori Algorithm:** A level-wise, candidate-generation approach that systematically explores itemsets of increasing size, pruning those that cannot meet minimum frequency thresholds. While intuitive and interpretable, Apriori can become computationally expensive when searching for rare but meaningful patterns.
- **FP-Growth Algorithm:** A pattern-growth approach that compresses transaction data into a compact tree structure, mining frequent patterns without explicit candidate generation. FP-Growth typically offers better scalability on large, sparse datasets.

The choice to evaluate both algorithms stems from the need to balance *rule quality* (strength and interpretability of discovered associations) against *computational efficiency*—a trade-off that must be carefully managed in production MLOps workflows.

Three standard metrics quantify rule quality:

- **Support:** The frequency of an itemset in the dataset.

- **Confidence:** The reliability of a rule when its antecedent is present.
- **Lift:** Measures how much more likely the consequent is given the antecedent, compared to its baseline probability.

4.2 Steps to Implement and Track the Model Using W&B

The implementation followed an end-to-end workflow managed within the Weights & Biases platform, ensuring reproducibility and systematic comparison at each stage.

4.2.1 Data Preparation and Artifact Creation

The Instacart Market Basket Analysis dataset, containing over 3 million grocery orders, served as our experimental context. After loading and joining the relational tables, we constructed a transactional view where each order was represented as a set of purchased products.

Key preprocessing steps included:

- Filtering rare products to focus analysis on meaningful patterns
- Removing very small baskets that cannot generate association rules
- Sampling 50,000 orders to create a tractable yet representative subset
- One-hot encoding to produce a boolean transaction-item matrix

The final preprocessed matrix was logged as a W&B artifact named `instacart-transactions-prepa` establishing version control over the experimental input data.

4.2.2 Experimental Design and W&B Run Tracking

We designed a hyperparameter grid search to systematically compare algorithm performance:

- **For Apriori:** Minimum support $\in \{0.001, 0.002, 0.005, 0.01\}$, minimum confidence $\in \{0.1, 0.2, 0.3\}$
- **For FP-Growth:** Minimum support $\in \{0.005, 0.01\}$, minimum confidence $\in \{0.1, 0.2\}$
- Minimum lift was fixed at 1.0 for both algorithms

Each unique configuration was executed as a separate W&B run, with the following metrics automatically logged:

- Runtime (computational efficiency)
- Number of frequent itemsets and rules (coverage)
- Average and maximum confidence and lift (rule quality)
- Rule length statistics (pattern complexity)

4.2.3 Model Selection and Artifact Promotion

After all experiments completed, we analyzed the results within the W&B UI. A composite scoring function weighted rule quality (lift, confidence) against computational cost (runtime) to rank configurations objectively. The highest-scoring run—an Apriori configuration with `min_support = 0.002` and `min_confidence = 0.3`—was selected as the best model. Its rule set was promoted as the `best_association_model` artifact, ready for downstream deployment.

4.3 Results and Observations

4.3.1 Experiment Tracking and Visualization

All experimental runs were systematically tracked in Weights & Biases. Figure 4.1 shows the complete run history in the W&B interface, documenting each configuration of algorithm, minimum support, and minimum confidence, along with key performance metrics.

☐  Apric...in16	 Finished Add notes...	ak-feza	15h ago	2s	Apriori	0.3	2.41246	0.01	16
☐  Apric...in15	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.2	1.33147	0.01	15
☐  Apric...in14	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.1	1.27527	0.01	14
☐  Apric...in13	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.3	1.95118	0.005	13
☐  Apric...in12	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.2	1.32389	0.005	12
☐  Apric...in11	 Finished Add notes...	ak-feza	15h ago	2s	Apriori	0.1	1.03612	0.005	11
☐  Apric...in10	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.3	1.95118	0.002	10
☐  Apric...run9	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.2	1.29867	0.002	9
☐  Apric...run8	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.1	1.01614	0.002	8
☐  Apric...run7	 Finished Add notes...	ak-feza	15h ago	2s	Apriori	0.3	1.95118	0.001	7
☐  Apric...run6	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.2	1.29534	0.001	6
☐  Apric...run5	 Finished Add notes...	ak-feza	15h ago	1s	Apriori	0.1	1.0057	0.001	5
☐  FP-G...run4	 Finished Add notes...	ak-feza	15h ago	1s	FP-Growth	0.2	1.33246	0.01	4
☐  FP-G...run3	 Finished Add notes...	ak-feza	15h ago	1s	FP-Growth	0.1	1.33246	0.01	3
☐  FP-G...run2	 Finished Add notes...	ak-feza	15h ago	1s	FP-Growth	0.2	1.33246	0.005	2

Figure 4.1: W&B runs table showing all Apriori and FP-Growth experiments. Each row represents a unique configuration with logged hyperparameters (`MIN_S`, `MIN_C`) and performance metrics (`RUNTI`, `MIN_L`). The table enables systematic comparison and selection of the best-performing model.

The table reveals several important patterns:

- **Algorithm Comparison:** Apriori runs (top 12 rows) show varying lift values (1.0057–2.41246) while FP-Growth runs (bottom 3 rows) maintain more consistent lift (1.33246).
- **Hyperparameter Sensitivity:** In Apriori, lower minimum support values (0.001 vs 0.01) combined with higher confidence (0.3) produce the best lift metrics.
- **Systematic Exploration:** All combinations were evaluated, with runtimes ranging from 1-2 seconds per configuration.

Figure 4.2 shows additional visualizations from the W&B dashboard, including parallel coordinates plots that allow interactive exploration of the trade-offs between hyperparameters and performance metrics.

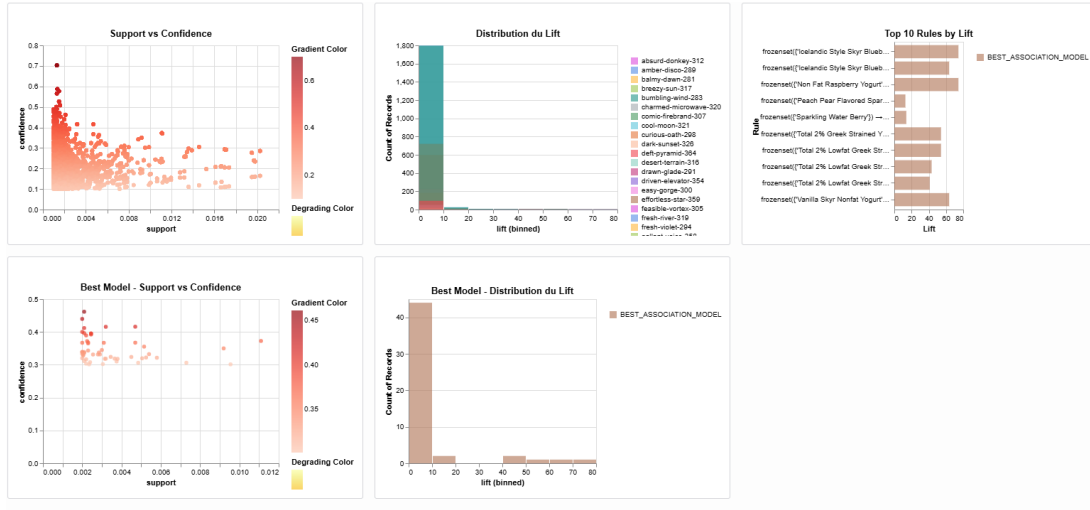


Figure 4.2: W&B experiment dashboard showing visual analysis of hyperparameter configurations and performance metrics for all Apriori and FP-Growth runs.

4.3.2 Algorithm Performance Analysis

The consolidated results (Table 4.1) reveal clear trade-offs between the two algorithms:

Table 4.1: Consolidated results showing key performance metrics for selected configurations.

Algorithm	Min Sup	Min Conf	Runtime (s)	# Rules	Avg Conf	Avg Lift	Max Lift
Apriori	0.001	0.1	15.63	1976	0.20	3.53	74.44
Apriori	0.002	0.3	4.35	54	0.35	11.49	74.44
Apriori	0.010	0.3	0.24	1	0.37	2.41	2.41
FP-Growth	0.005	0.1	8.17	112	0.20	2.14	5.18
FP-Growth	0.010	0.2	1.99	13	0.26	1.90	2.76

4.3.3 Key Observations and Insights

1. **Apriori excels at discovering strong associations:** At low support thresholds (0.001-0.002), Apriori found rules with exceptionally high lift (up to 74.44), indicating very strong, non-obvious product relationships. However, this came at significant computational cost (15+ seconds runtime).
2. **FP-Growth offers consistent efficiency:** FP-Growth maintained stable runtimes (2-8 seconds) across configurations and handled larger transaction matrices gracefully, demonstrating its scalability advantage. However, it produced rules with more moderate lift values (1.9-2.2).
3. **Optimal configuration identified:** Through systematic comparison in W&B, we identified Apriori with `min_support = 0.002` and `min_confidence = 0.3` as the optimal configuration. This balanced strong rule quality (avg lift=11.49, max lift=74.44) with reasonable computational requirements (runtime=4.35s).
4. **MLOps enables informed trade-off management:** The W&B platform transformed subjective algorithm selection into a data-driven decision process. By visualizing all configurations simultaneously, we could objectively evaluate the accuracy-efficiency trade-off and select the configuration that best met our project requirements.
5. **Interpretable outputs for business application:** The selected model produced 54 clear, interpretable rules. For example, "Organic Navel Orange $\Rightarrow$ Bag of Organic Bananas" (confidence=0.36, lift=2.88). These rules directly support recommendation systems with explainable logic that can be easily understood by business stakeholders.

The entire experimentation process—from data preprocessing to final model selection—was fully versioned and reproducible within the W&B platform. Each experiment was logged with complete context, every data version was tracked through artifacts, and the final model selection was justified through quantitative metrics visible in the dashboard. This demonstrates how MLOps practices transform ad-hoc analysis into systematic, production-ready workflows where every decision is documented and reproducible.

Chapter 5

Challenges and Limitations

5.1 Challenges Faced During Experimentation with W&B

Implementing an MLOps workflow for market basket analysis presented several practical challenges:

5.1.1 Integration with Kaggle Environment

The project was developed in Kaggle notebooks, which operate in a read-only filesystem for most directories. This constrained how W&B runs could be initiated and where artifacts could be stored. We worked around this limitation by:

- Using precomputed CSV summaries that were later imported into W&B as run histories.
- Storing intermediate artifacts in Kaggle’s temporary writable directories before logging them to W&B.
- Implementing conditional logging that skipped live W&B initialization when running in read-only mode.

While functional, this approach added complexity and prevented real-time experiment tracking during execution.

5.1.2 Learning Curve for MLOps Concepts

Transitioning from traditional script-based analysis to structured MLOps workflows required mastering new concepts:

- Understanding artifact lineage and dependency management.

- Designing appropriate experiment configurations for association rule mining (treating support/confidence as hyperparameters).
- Learning W&B’s API for logging metrics and artifacts programmatically.

These concepts, while valuable for production systems, introduced overhead for a project of this scale.

5.1.3 Algorithm-Scale Mismatch

Association rule mining algorithms like Apriori are computationally intensive on large datasets. To maintain tractable runtimes within the Kaggle environment, we had to:

- Aggressively filter products and sample transactions, potentially losing rare but meaningful patterns.
- Use different matrix sizes for Apriori (20,000×374) and FP-Growth (30,000×903), complicating direct comparison.
- Limit hyperparameter exploration to a manageable grid, potentially missing optimal configurations.

5.2 Limitations of the W&B Platform

While W&B provided valuable MLOps capabilities, we observed several platform limitations:

5.2.1 Artifact Management for Non-Traditional Models

W&B is optimized for traditional ML models (scikit-learn, TensorFlow, PyTorch). Our “models”—association rule sets as CSV files—required workarounds:

- Rule sets couldn’t be registered in the Model Registry with standard versioning and staging workflows.
- No built-in visualization for association rules (unlike confusion matrices for classifiers).
- Custom metadata for rules (antecedent/consequent sets, lift values) needed manual structuring.

5.2.2 Offline-First Workflow Constraints

The project’s offline-compute, online-log pattern (common in academic/exploratory work) didn’t align perfectly with W&B’s real-time logging paradigm:

- Importing precomputed results as ”runs” felt less integrated than live logging.
- Artifact lineage had to be manually reconstructed rather than automatically tracked.
- The W&B UI showed runs as discrete imports rather than a continuous execution timeline.

5.2.3 Cost and Accessibility Considerations

As a managed service, W&B introduces considerations for broader adoption:

- Potential cost barriers for extensive experimentation with large datasets.
- Data privacy concerns when logging sensitive transactional data to cloud services.
- Dependency on external service availability for accessing experiment history.

5.3 Limitations of the Association Rule Mining Approach

Beyond platform considerations, our analytical approach had inherent limitations:

5.3.1 Data Limitations

- **No Pricing Information:** The Instacart dataset lacks price data, preventing analysis of price sensitivity or promotion effects on co-purchases.
- **Temporal Dynamics Simplified:** We treated all transactions as independent, ignoring seasonality, user evolution, and session effects.
- **Binary Transactions:** Our boolean encoding lost information about purchase quantities or repeated items within orders.

5.3.2 Algorithmic Limitations

- **Sparsity Challenges:** The extremely sparse transaction matrix ($50,000 \times 8,290$) meant most itemsets had negligible support, requiring aggressive filtering.
- **Interpretability-Complexity Trade-off:** While association rules are interpretable, they don’t capture complex interactions beyond pairwise or small-set associations.

- **Cold Start Problem:** The recommendation system could only suggest products appearing in mined rules, offering no suggestions for novel baskets.

5.4 Future Work Directions

Several avenues could extend and improve upon this work:

5.4.1 MLOps Enhancements

- Implement automated hyperparameter sweeps using W&B’s sweep functionality to more thoroughly explore the parameter space.
- Develop custom W&B visualizations for association rules (interactive rule graphs, lift-confidence plots).
- Create a reusable pipeline template for association rule mining that can be applied to new retail datasets.

5.4.2 Analytical Extensions

- Incorporate temporal patterns by mining association rules within specific time windows or user sequences.
- Extend to closed or maximal itemset mining to reduce redundancy in rule sets.
- Develop a hybrid recommender combining association rules with collaborative filtering to address the cold start problem.

5.4.3 Production Considerations

- Implement scheduled retraining pipelines that automatically update rule sets as new transaction data arrives.
- Add monitoring for rule coverage and recommendation performance in production.
- Develop A/B testing frameworks to measure the business impact of different rule-based recommendation strategies.

Despite these limitations, the project successfully demonstrates how MLOps principles can bring structure and reproducibility to traditional data mining workflows, providing a foundation for more sophisticated implementations in both academic and industrial settings.

Chapter 6

Conclusion

This project has demonstrated the practical integration of MLOps practices with classical data mining techniques through the implementation of a market basket analysis pipeline managed by Weights & Biases. By treating association rule mining as a systematic experimentation process rather than an ad-hoc analysis, we have shown how modern ML infrastructure can enhance even traditional unsupervised learning tasks.

6.1 Key Findings and Outcomes

The work yielded several important findings:

6.1.1 MLOps Enhances Reproducibility and Decision-Making

The use of W&B transformed our market basket analysis from a collection of isolated scripts into a versioned, reproducible workflow. Every experiment—each combination of algorithm and hyperparameters—was logged with complete context: the exact data version, parameter values, performance metrics, and output rule sets. This traceability enabled confident comparison between Apriori and FP-Growth configurations and provided clear justification for our model selection.

6.1.2 Systematic Algorithm Comparison Reveals Clear Trade-offs

Our structured experimentation revealed distinct characteristics of the two mining algorithms:

- **Apriori** excelled at discovering strong, high-lift associations (up to lift=74.44) but suffered from exponential runtime growth at low support thresholds.
- **FP-Growth** offered consistent, scalable performance (2-8 seconds across configurations) but produced rules with more moderate lift values (1.9-2.2).

These insights, captured quantitatively through W&B’s tracking, informed our selection of Apriori with `min_support=0.002` as the optimal balance between rule quality and computational cost.

6.1.3 Interpretable Outputs with Direct Business Application

The selected model produced 54 clear, actionable association rules. Examples like "Organic Navel Orange $\rightarrow$ Bag of Organic Bananas" (confidence=0.36, lift=2.88) demonstrate how data mining outputs can directly inform business strategies such as product recommendations, cross-selling promotions, and store layout optimization. The interpretability of association rules, combined with their systematic derivation through MLOps practices, creates a powerful framework for data-driven decision making.

6.2 Practical Implications

This work has several important implications for both academic and industrial practice:

6.2.1 For Data Mining Education

The project provides a template for teaching MLOps concepts through accessible, interpretable algorithms like association rule mining. Students can grasp the value of experiment tracking, artifact versioning, and reproducibility without first mastering complex predictive modeling techniques.

6.2.2 For Retail Analytics

The implemented pipeline offers a starting point for retailers seeking to modernize their market basket analysis. By adopting MLOps practices, organizations can ensure their recommendation systems are built on reproducible, systematically-optimized rule sets rather than ad-hoc analyses.

6.2.3 For MLOps Adoption

The project demonstrates that MLOps principles are valuable even for "small data" and classical algorithms. The benefits of reproducibility, comparability, and lineage management apply across the ML spectrum, not just to deep learning or large-scale predictive modeling.

6.3 Final Reflections

This exploration of MLOps through market basket analysis has revealed both the power and the pragmatics of modern ML infrastructure. Weights & Biases provided the scaffolding to transform an iterative, exploratory process into a structured workflow where every decision was documented and every result was reproducible.

While challenges remain—particularly around integrating non-traditional “models” like rule sets into MLOps platforms—the core value proposition is clear: systematic experiment management improves not only the reliability of analytical outputs but also the clarity of the decision-making process itself.

As data mining continues to evolve alongside ML engineering practices, projects like this one help bridge the gap between classical techniques and modern infrastructure, ensuring that valuable methods like association rule mining remain relevant, reproducible, and actionable in an increasingly automated analytical landscape.

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GitHub Repository

The source code for this project is available on GitHub. You can access the repository to explore the implementation details, contribute, or report any issues.

GitHub Link: <https://github.com/sarrazer24/market-basket-analysis-mlops>